

RABBIT Bipedal Robot

A MATLAB Simulation and Hybrid Zero Dynamics Gait Framework

Technical Documentation

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Abstract

This document describes a modular MATLAB framework for modeling, simulation, control, and trajectory optimization of the RABBIT planar five-link underactuated biped. The centerpiece is a working Hybrid Zero Dynamics (HZD) gait-optimization pipeline that produces dynamically feasible, periodic walking gaits and animates them. We cover the robot model, the continuous and hybrid dynamics, the HZD control formulation, the nonlinear program solved with `fmincon`, the software architecture and data flow, a per-folder file reference, and reproducibility notes. Deviations between the documented and the as-built code (broken legacy entry points, orphaned files) are flagged explicitly.

1 Overview

RABBIT is a planar, five-link, *underactuated* biped: a torso plus two legs, each with a hip and a knee. Walking is modelled as a hybrid dynamical system — continuous single-support (swing) dynamics punctuated by a discrete impact when the swing foot strikes the ground. A gait is designed by imposing *virtual constraints* (B-spline reference trajectories for the actuated joints) and choosing their coefficients, together with the periodic initial state, so that the closed-loop system admits a stable periodic orbit while minimizing actuator effort.

Dependencies. MATLAB with the **Symbolic Math Toolbox** (only to *regenerate* the dynamics; the generated files are committed) and the **Optimization Toolbox** (`fmincon`, SQP — required for the gait pipeline).

Working entry point.

<code>startup</code>	% add all folders to the path
<code>cd Trajectory_Optimization</code>	
<code>rabbit_hzd_trajectory_optimization</code>	% optimize one gait, inspect, animate

A gait library across walking speeds is built with `build_gait_library.m`. See Section ?? for legacy entry points that no longer run.

2 Robot model

The generalized coordinates (7 DOF) are

$$q = [p_x \ p_z \ q_t \ q_1 \ q_2 \ q_3 \ q_4]^\top, \quad x = \begin{bmatrix} q \\ \dot{q} \end{bmatrix} \in \mathbb{R}^{14}. \quad (1)$$

symbol	meaning	actuated
p_x, p_z	torso base position (world frame)	no
q_t	torso pitch angle	no
q_1, q_2	stance-leg hip, knee	yes
q_3, q_4	swing-leg hip, knee	yes

Table 1: Generalized coordinates. Four actuators drive $q_1 \dots q_4$; the floating base (p_x, p_z, q_t) is unactuated, giving one degree of underactuation in single support.

The input map (`Dynamics/input_matrix.m`) is the constant

$$B = \begin{bmatrix} 0_{3 \times 4} \\ I_4 \end{bmatrix} \in \mathbb{R}^{7 \times 4}. \quad (2)$$

Physical parameters. There is no separate model/parameter file; the physical parameters are baked into the symbolic derivation (`Dynamics/rabbit_energy_model_generalized_Lagrange.m`, nested `Mass_Properties`): torso 10 kg / 0.75 m; each leg link (thigh, shank) 5 kg / 0.5 m; gravity $g = 9.8062 \text{ m/s}^2$.

Sign convention (important). The world-frame Z axis is *up-positive* (ground = 0, above ground > 0), but p_z is stored *down-positive* in the generalized coordinates, so the world-frame hip height is $-p_z$. All kinematic helpers (`P_st`, `P_sw`, `Tt`, `get_body_points`) return world-frame, up-positive heights. Mixing the two conventions has historically caused sign bugs.

3 Continuous dynamics

The unconstrained equations of motion are

$$M(q) \ddot{q} + V(q, \dot{q}) + G(q) = B u, \quad (3)$$

with $M(q)$ the mass/inertia matrix (`Dynamics/M.m`), $V(q, \dot{q})$ the Coriolis/centrifugal vector (`Dynamics/V.m`, called as `V([q; dq])`), and $G(q)$ the gravity vector (`Dynamics/G.m`).

Single-support (constrained) dynamics. During swing the stance foot is pinned by the holonomic contact constraint $p_{\text{st}}(q) = \text{const}$, i.e. at the acceleration level

$$J_{\text{st}}(q) \ddot{q} + \dot{J}_{\text{st}} \dot{q} = 0, \quad J_{\text{st}} = \frac{\partial p_{\text{st}}}{\partial q} \in \mathbb{R}^{2 \times 7}. \quad (4)$$

`Dynamics/rabbit_constrained_dynamics.m` solves the resulting saddle-point (KKT) system for the accelerations and the contact wrench λ :

$$\begin{bmatrix} M & -J_{\text{st}}^\top \\ J_{\text{st}} & 0 \end{bmatrix} \begin{bmatrix} \ddot{q} \\ \lambda \end{bmatrix} = \begin{bmatrix} B u - V - G \\ -\dot{J}_{\text{st}} \dot{q} \end{bmatrix}. \quad (5)$$

4 Hybrid model

Position is continuous across impact; only velocity jumps.

- **Guard / event** (`Contact/rabbit_impact_event.m`): the swing-foot height $p_{\text{sw},z}(q)$ crosses zero from above.

- **Impact map** (`Contact/rabbit_impact_map.m`): momentum-conserving reset enforcing zero post-impact swing-foot velocity,

$$\begin{bmatrix} M & -J_{\text{sw}}^\top \\ J_{\text{sw}} & 0 \end{bmatrix} \begin{bmatrix} \dot{q}^+ \\ \lambda \end{bmatrix} = \begin{bmatrix} M \dot{q}^- \\ 0 \end{bmatrix}. \quad (6)$$

- **Reset map** (`Reset_Map/rabbit_reset_map.m`): relabels stance/swing legs (`relabel_state.m`) and re-plants the new stance foot on the ground ($p_z \leftarrow p_{\text{st},z}$). The horizontal coordinate is intentionally not shifted, so the robot walks forward continuously in the world frame.

5 Phase variable and Hybrid Zero Dynamics

Phase variable (linear in q). The gait “clock” is the absolute stance-leg angle, *linear* in q :

$$\theta(q) = q_t + q_1 + \frac{1}{2}q_2 = c q, \quad c = \begin{bmatrix} 0 & 0 & 1 & 1 & \frac{1}{2} & 0 & 0 \end{bmatrix}. \quad (7)$$

This is implemented in `Controller/theta_of_q.m`; its gradient $\partial\theta/\partial q = c$ is the constant returned by `Controller/dtheta_dq_of.m`. Because both leg links have length $\frac{1}{2}$, the geometric hip-to-foot angle `atan2` equals (??) exactly for all physical poses ($|q_2| < \pi$); the linear form is preferred because *it cannot wrap at $\pm\pi$* and matches Westervelt’s Hypothesis HH6, on which the HZD decoupling analysis relies.

Virtual constraints. The normalized phase is

$$s = \frac{\theta - \theta^-}{\theta^+ - \theta^-} \in [0, 1], \quad (8)$$

where θ^- is the phase at step start and θ^+ (`p.theta_plus`) at touchdown. The four actuated joints track a clamped cubic B-spline reference $y_d(s)$,

$$y(q) = q_{4:7} - y_d(s), \quad u = -K_p y - K_d \dot{y}, \quad (9)$$

with $K_p = 400$, $K_d = 40$. The B-spline is evaluated by `bspline_eval.m` (basis functions from `BSpline.m` / `BSpline_derivative.m`); the whole closed loop is assembled in `Dynamics/hzd_closed_loop_ode.m` whose augmented state $\xi = [x; \text{torque cost}]$ integrates the running cost alongside the dynamics.

6 Trajectory optimization (the nonlinear program)

Decision vector.

$$z = \begin{bmatrix} \text{vec}(\text{coeffs}) \\ x_{\text{start}} \end{bmatrix} \in \mathbb{R}^{n_u n_c + 14} = \mathbb{R}^{38}, \quad (10)$$

with $n_u = 4$ actuated outputs and $n_c = 6$ control points each (`unpack_z.m`).

Cost (`hzd_cost.m`): squared torque per unit step length (Westervelt eq. 6.43),

$$J(z) = \frac{1}{L_{\text{step}}} \int_0^{T_{\text{step}}} \|u(t)\|_2^2 dt, \quad (11)$$

where L_{step} is the stance-to-swing foot distance at impact. Normalizing by L_{step} removes the degenerate incentive to “step in place”.

Constraints (`hzd_constraints.m`). Equalities ($\text{ceq} \in \mathbb{R}^{18}$):

- one-step periodicity, $x_{\text{next}}(2:14) = x_{\text{start}}(2:14)$, with $x_{\text{next}} = \Delta_{\text{reset}}(\Delta_{\text{impact}}(x_{\text{end}}))$ (13);
- stance foot on ground at $t = 0$ (1);
- stance foot zero velocity, $J_{\text{st}}\dot{q}_{\text{start}} = 0$ (2);
- touchdown phase, $\theta(q_{\text{end}}) = \theta^+$ (1);
- **NEC1** average walking rate, $L_{\text{step}}/T_{\text{step}} = v_{\text{des}}$ (1).

Inequalities ($c \in \mathbb{R}^4$, $c \leq 0$): no ground penetration; mid-step swing-foot clearance (**NIC3**); step-length floor; and orbital stability (Section ??, gated behind `p.enforce_stability`).

Solver (`hzd_problem_setup.m`). `fmincon` SQP with *central* finite differences (the event-triggered `ode45` makes forward differences noisy) and `ScaleProblem off` (so reported feasibility is directly comparable to the constraint tolerance). Cold starts from x_0 are unreliable; `build_gait_library.m` warm-starts each speed from the previously converged gait and marches v_{des} in small steps.

Diagnostics. `inspect_solution.m` prints and returns a full breakdown of any z : the θ sweep, step length/duration, walking speed, cost, every constraint residual, the worst violation, the within-step stance-foot drift, and the orbital-stability radius ρ (below).

7 Orbital stability: periodic is not the same as stable

This section is deliberately expanded for teaching, because it is the single most instructive point in the pipeline.

The trap. The periodicity equality makes x_{start} a *fixed point* of the step-to-step (Poincaré) map

$$x_{\text{next}} = P(x_{\text{start}}) := \Delta_{\text{reset}}(\Delta_{\text{impact}}(\varphi_{\text{step}}(x_{\text{start}}))), \quad (12)$$

where φ_{step} is one closed-loop swing phase. But *a fixed point can be unstable*. Satisfying periodicity to machine precision guarantees a limit cycle *exists*; it says nothing about whether the robot stays on it.

The diagnosis. Linearize (??) about the fixed point, $A = \partial P / \partial x$, and look at its spectral radius

$$\rho = \max_i |\lambda_i(A)|. \quad (13)$$

The gait is orbitally (asymptotically) stable iff $\rho < 1$. This is Westervelt’s stability requirement (informally, NEC5). For the reference seed gait obtained by enforcing periodicity *alone*, the numbers are stark:

max periodicity	ρ	open-loop $ \dot{q} $ over 6 steps
0.007 (excellent)	≈ 3 (unstable)	$6 \rightarrow 6 \rightarrow 19 \rightarrow 891 \rightarrow 949 \rightarrow 440$

The orbit is a near-perfect fixed point that the robot falls off after two steps. Every downstream symptom we observed — stance-foot drift growing across a step, Baumgarte constraint stabilization diverging, the Poincaré settling iteration blowing up, fragile speed continuation — is the same instability seen from a different angle: an orbit with $\rho > 1$ amplifies *every* perturbation, numerical or physical.

The remedy. Add $\rho < 1$ as an optimization constraint. `poincare_stability.m` builds A by central finite differences on the 13 periodic states (index 2:14; p_x is the translation gauge and is excluded),

$$A_{:,j} = \frac{P(x_{\text{start}} + h e_j) - P(x_{\text{start}} - h e_j)}{2h}, \quad j = 1, \dots, 13, \quad (14)$$

and returns $\rho = \max_i |\lambda_i(A)|$. `hzd_constraints.m` then adds the inequality $\rho - \rho_{\text{max}} \leq 0$ (with $\rho_{\text{max}} = 0.95$).

Cost and workflow. Each ρ evaluation costs $\approx 2 \times 13 = 26$ step simulations, and `fmincon` finite-differences the constraint over 38 variables — hundreds to thousands of simulations per iteration. The constraint is therefore **gated** behind `p.enforce_stability` (default `false`). The recommended workflow is two-phase:

1. solve a *periodic* gait with stability off (fast);
2. turn `p.enforce_stability = true` and re-solve *warm-started* from that gait to drive ρ below 1.

`inspect_solution.m` reports ρ on every call regardless of the gate, so stability is always visible even when it is not being enforced.

Analogy. Just as the *step-length floor* was the missing constraint that stopped the optimizer exploiting a degenerate “step in place”, the *stability bound* is the missing constraint that stops it returning a mathematically periodic but physically un-walkable orbit. In both cases the optimizer was doing exactly what it was told; the specification was incomplete.

8 Software architecture and data flow

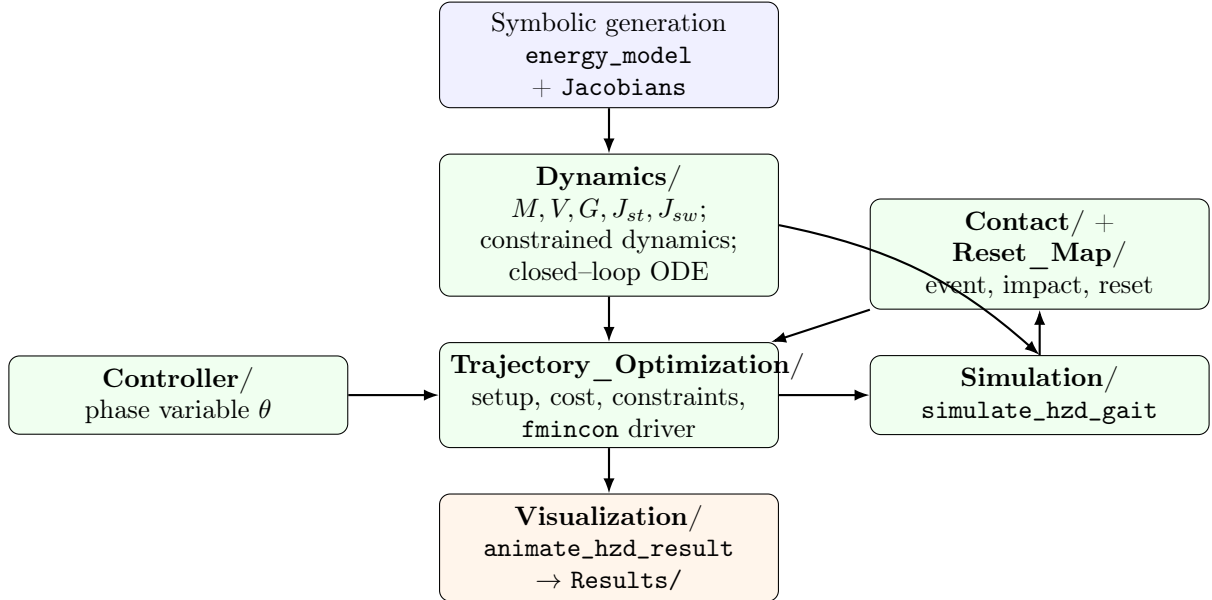


Figure 1: Pipeline: offline symbolic generation feeds the **Dynamics** layer; the optimizer repeatedly calls the single-step simulator (which integrates the closed-loop dynamics and detects impact) to evaluate the cost and constraints; the converged gait is animated and saved.

9 File reference

Dynamics/ (mostly Symbolic–Toolbox generated)

file	role
rabbit_energy_model_generator.m	Generalized Lagrange–Lagrange derivation; defines masses/inertias; produces $T_t, T_1\text{--}T_4, P_{st}, P_{sw}, M, DM, V, G$.
Jacobians.m	Generator: produces $J_{st}, J_{sw}, J_{dotdq_st}, J_{dotdq_sw}$.
Tt.m, T1–T4.m	Homogeneous transforms of torso / link frames, $T=T_t(q)$.
P_st.m, P_sw.m	Stance / swing foot position $[x; z]$, world frame.
M.m	Mass matrix $M(q)$ (7×7).
V.m	Coriolis/centrifugal vector, $V([q; dq])$.
G.m	Gravity vector $G(q)$.
DM.m	$\dot{M}$, $DM([q; dq])$ (used to form V).
J_st.m, J_sw.m	Foot Jacobians (2×7).
Jdotdq_st.m, Jdotdq_sw.m	$\dot{J}\dot{q}$ terms (2×1).
input_matrix.m	Constant actuation map B .
rabbit_constrained_dynamics.m	Single–support forward dynamics (KKT solve), $[ddq, lambda]=\dots(q, dq, u)$.
hzd_closed_loop_ode.m	Closed–loop swing RHS (B–spline + PD), augmented cost state.
rabbit_ode.m	Generic single–support RHS for an arbitrary controller.
rabbit_dynamics.m	Broken: calls nonexistent $D_matrix/C_vector/G_vector$.

Contact/ and Reset_Map/

file	role
rabbit_impact_event.m	ODE event: swing foot reaches ground.
impact_event_wrapper.m	Same event for the augmented HZD state ξ .
rabbit_impact_map.m	Momentum–conserving velocity reset, $x_{plus}=\dots(x_{minus})$.
relabel_state.m	Swap stance/swing joints and velocities.
rabbit_reset_map.m	Relabel + re–plant stance foot on ground.

Controller/

file	role
theta_of_q.m	Phase variable $\theta = q_t + q_1 + \frac{1}{2}q_2$.
dtheta_dq_of.m	Constant gradient c .
rabbit_controller.m	Broken: calls nonexistent <code>desired_gait</code> .

Trajectory_Optimization/ (main pipeline)

file	role
rabbit_hzd_trajectory_optimization.m	Entry point: single-gait solve, save, animate.
build_gait_library.m	Speed continuation; builds a <code>gait_library</code> struct.
hzd_problem_setup.m	Single source of p , bounds, <code>fmincon</code> options.
hzd_cost.m	Objective (torque ² per step length).
hzd_constraints.m	Equality + inequality constraints.
unpack_z.m	Split $z = [\text{coeffs}; x_{\text{start}}]$.
make_coeffs.m	Initial B-spline control points.
bspline_eval.m	Evaluate a spline output and its s -derivative.
BSpline.m, BSpline_derivative.m	Cox-de Boor basis and derivative.
inspect_solution.m	Full diagnostic breakdown of a solution (incl. ρ).
poincare_stability.m	Step-map Jacobian; returns spectral radius ρ (NEC5).
settle_initial_state.m	Poincaré iteration (disabled by default).

Simulation/, Utilities/, Visualization/

file	role
simulate_hzd_gait.m	One HZD step, summary outputs (status, penetration, clearance, T_{step}).
simulate_hzd_gait_full.m	One HZD step, full trajectory (for animation/drift).
simulate_one_step.m	Generic one step for an arbitrary controller.
simulate_n_steps.m	Chain generic steps with impact+reset.
get_body_points.m	World positions of key body points.
check_ground_validity.m	Ground/penetration/slip validation of a state.
animate_rabbit.m	Stick-figure animation + GIF export.
animate_hzd_result.m	Stitch n HZD steps and animate.

config/ and Test/

file	role
make_initial_state.m	Deterministic ground-valid start state (1 output).
make_random_initial_state.m	Rejection-sampled random start (unseeded).
config.m	Singleton: stepping-stone terrain + spline params.
init_bspline_params.m	Legacy B-spline controller config.
test_simulate_steps.m	Smoke test of the generic simulation path.

10 Reproducibility

- **Generated dynamics.** $T_t, T_1\text{--}T_4, P_{\text{st}}, P_{\text{sw}}, M, DM, V, G$ come from `rabbit_energy_model_generalized`. $J_{\text{st}}, J_{\text{sw}}, J_{\text{dotdq_st}}, J_{\text{dotdq_sw}}$ from `Jacobians.m`. *If any mass, length, inertia, or*

kinematic relation changes, these generator scripts must be re-run to regenerate the committed files. Do not hand-edit the generated files.

- **Saved results.** Results/hzd_result_<timestamp>.mat each store `z_opt`, `p`, `fval`, `exitflag`, `report`. The 2026-07-20_12-06-36 result is the reference periodic gait (≈ 0.355 m/s) used as the warm-start seed. `build_gait_library.m` writes `gait_library_<timestamp>.mat`.
- **Nondeterminism.** `make_random_initial_state.m` uses `randn` with no seed. The HZD pipeline instead uses a hard-coded x_0 , so its results are reproducible. Add `rng(seed)` before random sampling if determinism is needed there.
- **Orphaned cache.** `Trajectory_Optimization/Trajectory.mat` is committed but loaded by no code path; treat as unverified.
- **Simscape.** `Dynamics/Test_dynamics_Simscape.slx` is a standalone dynamics cross-check, not part of the runtime pipeline.

11 Known issues and broken legacy entry points

The following predate the HZD pipeline and **do not run as-is** — they call functions that no longer exist. They are retained for reference only.

- `main_demo.m`: calls `make_initial_state` expecting 3 outputs (it returns 1), `simulate_n_steps` with 4 arguments (it takes 3), and `animate_rabbit_stepping_stones` (does not exist).
- `Controller/rabbit_controller.m`: calls `desired_gait` / `desired_gait_velocity` (do not exist).
- `Dynamics/rabbit_dynamics.m`: calls `D_matrix` / `C_vector` / `G_vector` (do not exist; the real terms are M / V / G).

Open item: stance-foot constraint drift (a symptom). The stance foot is held only by the acceleration-level contact constraint, so `ode45` permits numerical drift of the foot within a step (~ 0.1 – 0.3 m; `inspect_solution.m` reports it). The natural fix is Baumgarte stabilization (`rabbit_constrained_dynamics.m` supports it via an optional `foot_ref`), but on the reference gait it *diverged* rather than helping. That is not a flaw in Baumgarte: as Section ?? shows, the seed orbit has $\rho \approx 3$, so it amplifies the very corrections Baumgarte injects. The drift is thus best read as *another symptom of orbital instability*; enforcing $\rho < 1$ addresses the root cause, after which constraint stabilization becomes well behaved.

Open item: enforcing stability is expensive. The $\rho < 1$ constraint is finite-difference heavy (Section ??). It is usable as a warm-started final phase but slow for a from-scratch solve; a reduced (2-dimensional hybrid zero dynamics) stability condition, or an analytic Poincaré Jacobian, would make it cheap. This is the natural next step for the pipeline, alongside the CLF-QP controller route (which stabilizes an open-loop-unstable orbit with feedback instead of redesigning it).

References

1. E. R. Westervelt, J. W. Grizzle, C. Chevallereau, J. H. Choi, B. Morris, *Feedback Control of Dynamic Bipedal Robot Locomotion*, CRC Press, 2007.